Chapter 8, Problem 3RQ

Problem

When a step input is applied to a second-order circuit, the final values of the circuit variables are found by:

(a) Replacing capacitors with closed circuits and inductors with open circuits.

(b) Replacing capacitors with open circuits and inductors with closed circuits.

(c) Doing neither of the above.

Step-by-step solution

Step 1 of 2

The final values of the circuit are found at $t = \infty$.

As the time, t and frequency, ω are inversely proportional. Calculate the impedances of the circuit at $\omega = 0 \text{ rad/s}$.

Calculate the impedance of the capacitor at $\omega = 0 \text{ rad/s}$ that at $t = \infty$.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(0)C} \\ &= \infty\ \Omega \end{aligned}$$

The impedance of the capacitor is infinity. As per Ohm’s law, current is inversely proportional to the impedance. So, as the impedance is infinity, the current through it is zero. It means capacitor acts as open circuit.

Step 2 of 2

Calculate the impedance of the inductor at $\omega = 0 \text{ rad/s}$ that at $t = \infty$.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(0)L \\ &= 0\ \Omega \end{aligned}$$

The impedance of the inductor is zero. As per Ohm’s law, current is inversely proportional to the impedance. So, as the impedance is zero, the current through it is infinity. It means inductor acts as short circuit.

Thus, the final values of the circuit variables are found by replacing the capacitors with open circuits and inductor with closed circuits (short circuit).

Thus, the correct answer is $\boxed{\text{b}}$.